

Uttkarsh Ruparel

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Education

Symbiosis Institute of Technology

Bachelor of Science in Computer Science (CGPA: 7.66 / 10)

July 2023 - June 2027

Pune, Maharashtra

Excellencia Junior College

Higher Secondary Education (Percentage: 85.5%)

July 2021 - May 2023

Secundrabad, Telangana

Projects

ESX — Stock Exchange Infrastructure | Go, Kafka, gRPC, PostgreSQL, Redis, AWS (EKS, MSK), Kubernetes

- Engineered a production-grade securities exchange — **8 microservices** (order gateway, settlement, matching engine, clearing house) over **Kafka** and **gRPC**, replicating core systems behind **NASDAQ**.
- Built a high-concurrency order gateway in **Go** sustaining **6,000+ RPS** at **sub-40ms latency** via **gRPC streaming** and a **Buffer & Drain pattern (500k-capacity buffered channels)** to decouple ingestion from matching, surviving **300% load spikes**.
- Implemented **atomic DvP settlement** with an **idempotent double-entry ledger**, preventing duplicate transactions via **unique ID tracking** and **atomic PostgreSQL upserts** — all movements net to zero.

Kprobe — Kernel Observability for Financial Systems | Rust, Aya, eBPF, Go, Python, Kafka, Neo4j, ClickHouse, OpenTelemetry, Jaeger

- Built a **Rust/Aya eBPF** probe attaching to **kernel tracepoints** at runtime — capturing **TCP RTT**, **CPU scheduling**, and **syscall latency** at **nanosecond resolution** with **zero application instrumentation**.
- Engineered a **Go/Kafka** stream processor that windows events into **100ms frames**, constructs a **directed causal graph** by PID/transaction co-occurrence within **50ms**, and classifies chains (SCHED_DELAY, MEMORY_PRESSURE, TCP_CHAIN) into **Neo4j** for **root cause attribution**.
- Layered **Isolation Forest** anomaly detection over **ClickHouse** kernel time-series on rolling **5-minute windows**; triggers an **LLM-powered RCA** pipeline that queries the **Neo4j** causal subgraph and generates incident reports via the **Gemini API**.

Hatch — Self-hosted Cloud Deployment Platform | Go (Gin), AWS (ECS, Fargate), RabbitMQ, Redis, PostgreSQL, Terraform, Docker

- Architected a self-hosted Render/Railway alternative on **AWS** — independent **Go** microservices for cloning, building and deploying containers on **ECS Fargate**, communicating through **RabbitMQ** with **real-time log streaming** via **Redis pub/sub** over **WebSockets**.
- Achieved **3,500 RPS** at **p99 < 75ms** with **zero failed requests** across **5,000 requests** (ApacheBench) and sustained **716 req/s** at **p95 < 9ms** under **50 concurrent VUs** (k6).

Technical Skills

Languages: Go, TypeScript, JavaScript, Python, Java, Rust

Backend & Systems: Node.js (Express, Fastify), Gin, FastAPI, REST, GraphQL, WebSockets, gRPC, Protocol Buffers, NGINX, Kong, PostgreSQL, MongoDB, Redis, Kafka, RabbitMQ, ClickHouse, Neo4j

Infrastructure & Observability: Docker, Kubernetes, AWS (MSK, ECR, SQS), Linux, Terraform, Helm, ArgoCD, CI/CD (GitHub Actions), Cloudflare, Prometheus, Grafana, OpenTelemetry, Jaeger, k6

Frontend & Testing: React, Next.js, Astro, Tailwind CSS

Foundations: System Design, Distributed Systems, Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Database Management Systems

Leadership Experience

President

2026 – Present

Google Developer Groups on Campus, Symbiosis Institute of Technology

Pune, India

- Leading a 40-member technical community; driving a project-first culture with active mentorship and community building.

Competitive Programming Co-Head

2025 – 2026

Google Developer Groups on Campus, Symbiosis Institute of Technology

Pune, India

- Conducted 20+ CP sessions and organized contests across the institute, growing participation to 100+ students.